# The Invariance Principle — problem 1.5

A. Engel — Problem-Solving Strategies · 1998

Around a circle, 5 ones and 4 zeros are arranged in any order. Then between any two equal digits, you write 0 and between different digits 1. Finally, the original digits are wiped out. If this process is repeated indefinitely, you can never get 9 zeros. Generalize !